

Circuits Reference Sheet (Full)

Units and SI Prefixes

$$\text{femto (f)} = 10^{-15}$$

$$\text{pico (p)} = 10^{-12}$$

$$\text{nano (n)} = 10^{-9}$$

$$\text{micro } (\mu) = 10^{-6}$$

$$\text{milli (m)} = 10^{-3}$$

$$\text{centi (c)} = 10^{-2}$$

$$\text{deci (d)} = 10^{-1}$$

$$\text{deca (da)} = 10^1$$

$$\text{hecto (h)} = 10^2$$

$$\text{kilo (k)} = 10^3$$

$$\text{mega (M)} = 10^6$$

$$\text{giga (G)} = 10^9$$

$$\text{tera (T)} = 10^{12}$$

$$\text{peta (P)} = 10^{15}$$

$$1e = 1.602 \times 10^{-19} \text{ C}$$

Energy

Energy storage: $1 \text{ J} = 1 \text{ W} \cdot \text{s}$, $1 \text{ Wh} = 3600 \text{ J}$

$$W = \int P dt = \int VI dt = VQ \quad \text{or} \quad P\Delta t \text{ (constant power)}$$

Units:

- W : energy in [J] or [Wh]
- P : power in [W]
- V : voltage in [V] or [J/C]
- I : current in [A] or [C/s]
- Q : charge capacity in [C] or [Ah]

Ex: Battery capacity

Battery life:

$$t = \frac{W}{P} \quad \frac{[\text{Wh}]}{[\text{W}]}$$

Charge time:

$$t = \frac{Q}{I_{\text{eff}}} \quad \frac{[\text{Ah}]}{[\text{A}]}, \quad I_{\text{eff}} = I_{\text{in}} \cdot \eta$$

Power

Power: $P = VI = I^2 R = \frac{V^2}{R}$

Resistance

Resistance: $R = \rho \frac{L}{A} \quad [\Omega]$

Conductance

Conductance: $G = \frac{1}{R} = \sigma \frac{A}{L} \quad [\text{S}]$

Resistors in Series and Parallel

Series:

$$R_{\text{eq}} = R_1 + R_2 + \cdots + R_n$$

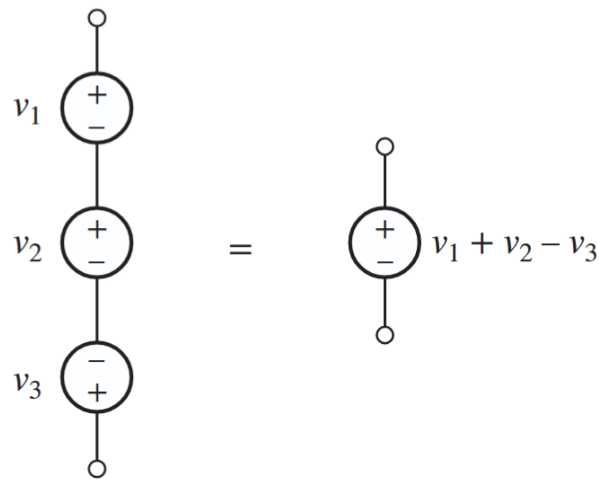
Parallel:

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}$$

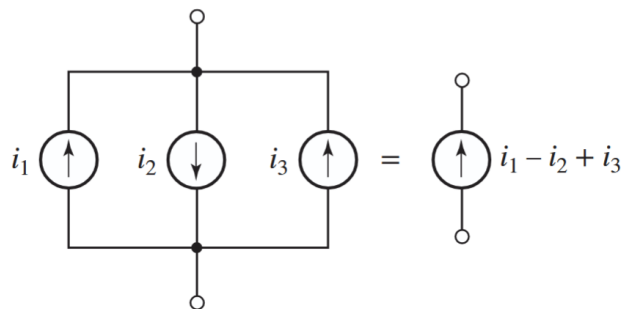
$$\Rightarrow R_{\text{eq}} = \frac{R_1 R_2}{R_1 + R_2}$$

*Combining Sources

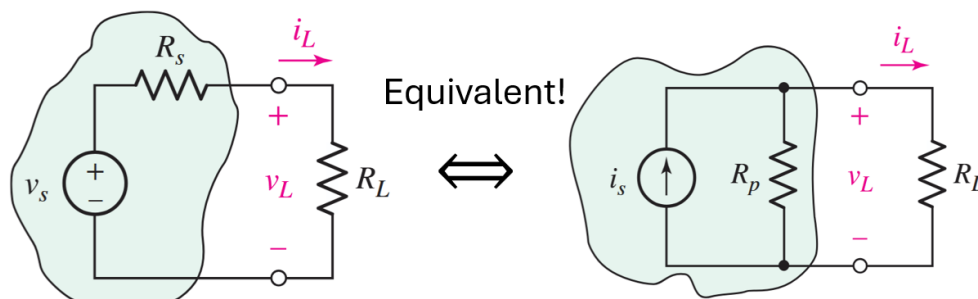
Voltage sources in series add



Current sources in parallel add



*Source Transformations



The circuits above are equivalent if

$$R_s = R_p \quad \text{AND} \quad V_s = i_s R_s$$

Nodal Analysis (KCL)

$$I_{R_1} = \frac{V_1 - V_2}{R_1} \quad (\text{voltage drop in the direction of current flow})$$

Supernode: Around *voltage source* ($V_3 - V_4 = V_{\text{source}}$)

Mesh Analysis (KVL)

$$V_{R_1} = I_{R_1} R_1$$

Supermesh: Loop around *current source* ($i_1 - i_2 = i_{\text{source}}$)

Voltage Division in Series

$$V_k = \frac{R_k}{R_1 + R_2 + \dots + R_n} V_{\text{source}}$$

Current Division in Parallel

$$I_k = \frac{\frac{1}{R_k}}{\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}} I_{\text{source}}$$

Superposition

Isolate each source, turn off all other source (voltage sources become short circuit, current sources become open circuit). Solve for circuit response from each source, then add responses together.

$$i_x = i'_x + i''_x, \quad v_x = v'_x + v''_x$$

Thevenin and Norton Equivalent Circuits

- Thevenin equivalent: Voltage source V_{Th} in series with resistance R_{Th}
- Norton equivalent: Current source i_N in parallel with resistor R_N
- They are source transformations of each other:

$$V_{Th} = i_N R_{Th}, \quad R_{Th} = R_N$$

Finding the Equivalent Source

- $V_{Th} = V_{oc}$ (open circuit voltage): disconnect the load R_L and solve for the voltage across the open terminals
- $i_N = i_{sc}$ (short circuit current): short the load (replace R_L with a wire) and solve for the current flowing through the short

Finding the equivalent resistance seen from the load R_L

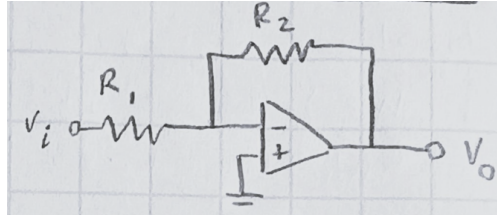
- If only independent sources: turn off all independent sources (voltage = short/wire, current = open/break). Sum resistances in remaining circuit
- If dependent source:
 - Find the open circuit voltage and short circuit current as above, then calculate $R_{Th} = \frac{V_{oc}}{i_{sc}}$
 - OR turn off independent sources, and apply a test voltage (V_{test} or 1V) and solve for the resulting current. Can also apply a test current (I_{test} or 1A) and solve for resulting voltage. Apply test voltage/current across the load terminals, and $R_{Th} = \frac{V_{test}}{I_{test}}$

Max power transfer condition:

$$R_L = R_{Th}$$

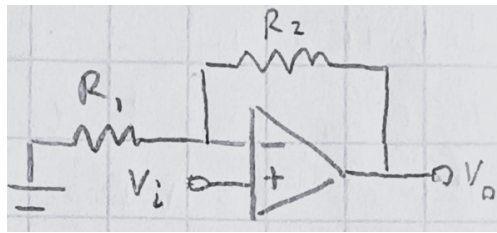
Operational Amplifiers

- Inverting Amplifier



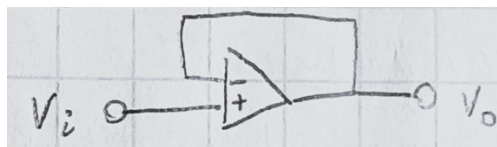
$$v_o = -\frac{R_2}{R_1}v_i$$

- Non-inverting Amplifier



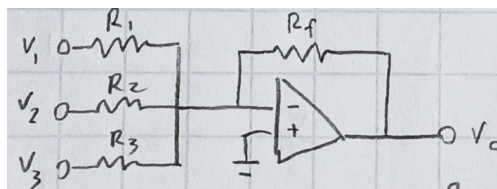
$$v_o = \left(1 + \frac{R_2}{R_1}\right)v_i$$

- Buffer (Voltage Follower)



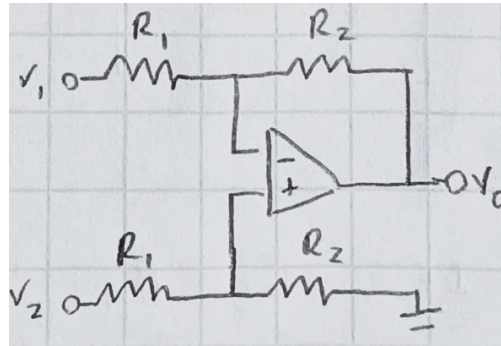
$$v_o = v_i$$

- Summer



$$v_o = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \frac{R_f}{R_3}v_3\right)$$

- **Difference Amplifier**



$$v_o = \frac{R_2}{R_1}(v_2 - v_1)$$

Energy Storage in Capacitors

Capacitance:

$$C = \frac{\epsilon_0 \epsilon_r A}{d} \quad (\text{parallel plate})$$

Where $\epsilon_0 = 8.854 \times 10^{-12}$ F/m is the permittivity of free space

Energy stored in a capacitor:

$$w = \frac{1}{2}CV^2 = \frac{q^2}{2C}$$

Summing capacitors in series and parallel:

- Series: $\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_n}$
- Parallel: $C_{eq} = C_1 + C_2 + \dots + C_n$

Properties of Capacitors

- A capacitor at steady state (DC) behaves like an open circuit
- *Voltage across a capacitor must be continuous:* $v_C(0^+) = v_C(0^-)$
- *Current through a capacitor can be discontinuous:* $i_C(0^+) \neq i_C(0^-)$

Energy Storage in Inductors

Inductance:

$$L = \frac{N^2 \mu A}{l} \quad (\text{solenoid})$$

Where $\mu = \mu_0 \mu_r$ is the permeability of the core material, and $\mu_0 = 4\pi \times 10^{-7}$ H/m is the permeability of free space

Energy stored in an inductor:

$$w = \frac{1}{2} L I^2$$

Summing inductors in series and parallel:

- Series: $L_{eq} = L_1 + L_2 + \dots + L_n$
- Parallel: $\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \dots + \frac{1}{L_n}$

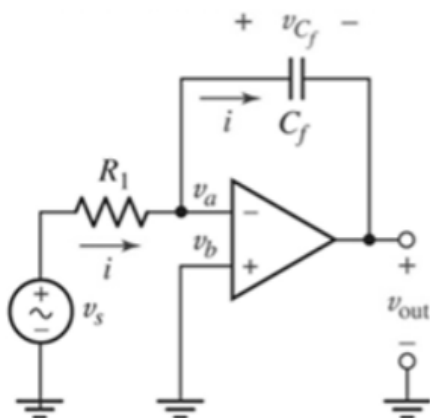
Properties of Inductors

- An inductor at steady state (DC) behaves like a short circuit
- *Current through an inductor must be continuous: $i_L(0^+) = i_L(0^-)$*
- *Voltage across an inductor can be discontinuous: $v_L(0^+) \neq v_L(0^-)$*

Relation	Resistor (R)	Capacitor (C)	Inductor (L)
$v - i :$	$v = iR$	$v = \frac{1}{C} \int_{t_0}^t i(\tau) d\tau + v(t_0)$	$v = L \frac{di}{dt}$
$i - v :$	$i = \frac{v}{R}$	$i = C \frac{dv}{dt}$	$i = \frac{1}{L} \int_{t_0}^t v(\tau) d\tau + i(t_0)$
P or $W :$	$p = i^2 R = \frac{v^2}{R}$	$w = \frac{1}{2} C v^2$	$w = \frac{1}{2} L i^2$
Series	$R_{eq} = R_1 + R_2$	$C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$	$L_{eq} = L_1 + L_2$
Parallel	$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$	$C_{eq} = C_1 + C_2$	$L_{eq} = \frac{L_1 L_2}{L_1 + L_2}$
At DC:	Same	Open circuit	Short circuit
Circuit variable that cannot change abruptly:	Not applicable	v	i

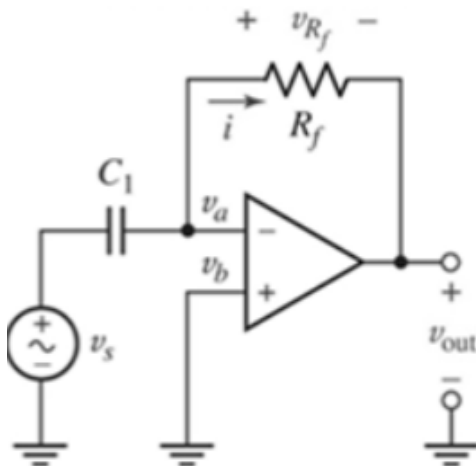
Operational Amplifiers with Energy Storage

- Integrator



$$v_o = -\frac{1}{R_1 C_f} \int_0^t v_s(t') dt' - v_{C_f}(0)$$

- Differentiator



$$v_o = -C_1 R_f \frac{dv_s}{dt}$$

First Order RC and RL Circuit Responses (DC)

Driven RC Circuit Step Response

1. With all independent sources zeroed out, simplify the circuit to determine R_{eq} and C_{eq} , and the time constant $\tau_{eq} = R_{eq}C_{eq}$.
2. Determine the initial condition $v(0^+)$ or $i(0^+)$ [recall that the capacitor voltage is continuous: $v_c(0^-) = v_c(0^+)$]
3. Determine the steady state condition $v(\infty)$ or $i(\infty)$.
4. The final response is given by:

$$\begin{aligned} v(t) &= v(\infty) + [v(0^+) - v(\infty)]e^{-t/\tau_{eq}} \\ i(t) &= i(\infty) + [i(0^+) - i(\infty)]e^{-t/\tau_{eq}} \end{aligned}$$

Driven RL Circuit Step Response

1. With all independent sources zeroed out, simplify the circuit to determine R_{eq} and L_{eq} , and the time constant $\tau_{eq} = L_{eq}/R_{eq}$.
2. Determine the initial condition $v(0^+)$ or $i(0^+)$ [recall the requirement that any inductor current $i_L(0^-) = i_L(0^+)$].
3. Determine the final condition $v(\infty)$ or $i(\infty)$.
4. The final response is given by

$$\begin{aligned} v(t) &= v(\infty) + [v(0^+) - v(\infty)]e^{-t/\tau_{eq}} \\ i(t) &= i(\infty) + [i(0^+) - i(\infty)]e^{-t/\tau_{eq}} \end{aligned}$$

Note: For a *natural response* (no forcing function), the steps above still apply, but the steady state values go to zero: $v(\infty) = 0$ and $i(\infty) = 0$.

Second Order RLC Circuit Responses (DC)

Defining Frequencies and Damping For a RLC circuit:

- Natural Frequency: $\omega_0 = \frac{1}{\sqrt{LC}}$
- Exponential damping coefficient: $\alpha = \frac{1}{2RC}$ (parallel RLC), $\alpha = \frac{R}{2L}$ (series RLC)
- Damping Ratio: $\zeta = \alpha/\omega_0$

Note: Overdamped: $\zeta > 1$, Critically damped: $\zeta = 1$, Underdamped: $\zeta < 1$.

Solving for the Time Domain Response:

Response Type	Condition	General Solution (for both $i(t)$ and $v(t)$)
Overdamped	$\alpha > \omega_0$	$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$
Critically Damped	$\alpha = \omega_0$	$i(t) = e^{-\alpha t}(A_1 t + A_2)$
Underdamped	$\alpha < \omega_0$	$i(t) = e^{-\alpha t}(B_1 \cos \omega_d t + B_2 \sin \omega_d t), \quad \omega_d = \sqrt{\omega_0^2 - \alpha^2}$

Note: The exponents s_1 and s_2 are simply $-\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$. Solve for the constants A_1 , A_2 , B_1 , and B_2 using the initial conditions of the problem.

In the case of a DRIVEN RLC circuit, the total response is the sum of the natural response and the forced response (steady state value for i or v).

Second Order RLC Circuit Responses (Sinusoidal Steady State)

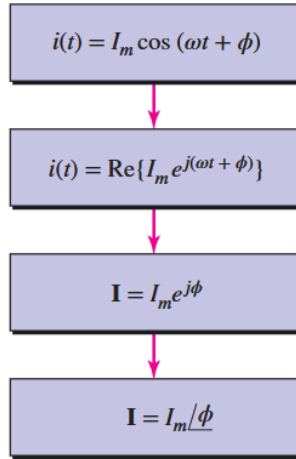
- A sinusoidal forcing function always results in a sinusoidal forced response of the same frequency in a linear circuit. This means that if the input is a sine then the output is a sine of the same frequency, but possibly different amplitude and phase.
- We can rewrite sinusoidal functions as the real and imaginary part of a complex exponential through Euler's identity

$$e^{j\omega t} = \cos(\omega t) + j \sin(\omega t)$$

$$\cos(\omega t + \phi) = \text{Re}\{e^{j(\omega t + \phi)}\}$$

$$\sin(\omega t + \theta) = \text{Im}\{e^{j(\omega t + \theta)}\}$$

- Since all sinusoidal functions have the same frequency, we can use phasor notation, suppressing the time dependence term $e^{j\omega t}$, resulting in a phasor.



Note that $\mathbf{I}$ represent complex numbers, and we will take the real or imaginary part of the resulting phasor to get to the actual time domain response.

Additionally, differentiation and integration in the time domain correspond to multiplication and division by $j\omega$ in the phasor domain, respectively.

- $\mathbf{V}$ and $\mathbf{I}$ relationships:

Time Domain		Frequency Domain
	$v = Ri$	$\mathbf{V} = R\mathbf{I}$
	$v = L \frac{di}{dt}$	$\mathbf{V} = j\omega L \mathbf{I}$
	$v = \frac{1}{C} \int i dt$	$\mathbf{V} = \frac{1}{j\omega C} \mathbf{I}$

We can use these complex valued currents and voltages with KCL and KVL as we normally would to analyze circuits, and then take the real or imaginary part of the resulting phasor to get to the actual time domain response.

- We define **impedance** to be the relationship between the complex voltage and current in a circuit element:

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}}$$

These impedances sum in series and parallel just like resistances and can also be used in nodal and mesh analysis using the relationship $\mathbf{V} = \mathbf{IZ}$ (just like Ohm's law).

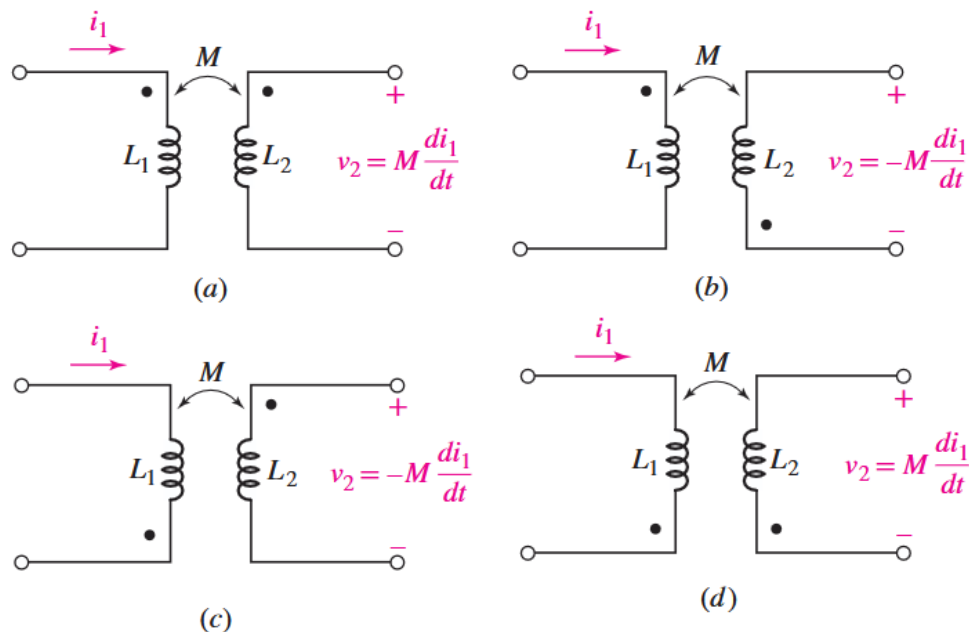
- Since $\mathbf{Z}$ is a complex number, we can express it either in rectangular form $\mathbf{Z} = R + jX$ or in polar form $\mathbf{Z} = |\mathbf{Z}|e^{j\theta}$, where R is the real part of the impedance (resistance), and X is the imaginary part of the impedance (reactance).

- Although we have complex voltages and currents for sinusoidal sources, because the circuit is linear, **superposition**, **source transformations**, and **Thevenin/Norton equivalents** still apply in the phasor domain. Just remember to use impedances instead of resistances.

Coupled Coils

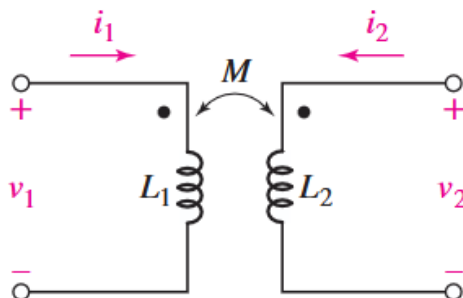
- Dot Convention for Mutual Inductance:**

- A current entering the *dotted* terminal of one coil produces an open-circuit voltage with a positive voltage reference at the *dotted* terminal of the second coil.
- A current entering the *undotted* terminal of one coil provides a voltage that is positively sensed at the *undotted* terminal of the second coil



Note that **self-induction** exists if the current in the right loop (i_2) is non-zero and would cause back emf.

- Total energy in coupled inductors:



$$w(t) = \frac{1}{2}L_1[i_1(t)]^2 + \frac{1}{2}L_2[i_2(t)]^2 + M[i_1(t)][i_2(t)]$$

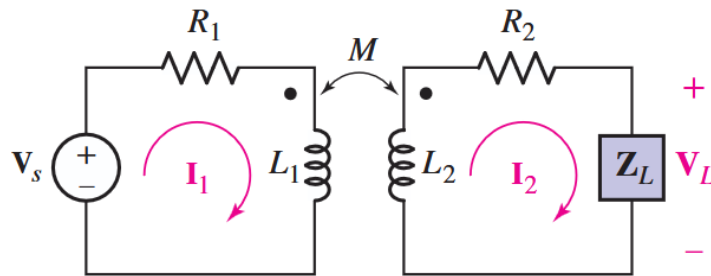
Limit on M:

$$M \leq \sqrt{L_1 L_2}$$

Coupling coefficient:

$$k = \frac{M}{\sqrt{L_1 L_2}} \implies 0 \leq k \leq 1$$

- Linear transformer and reflected impedance:



The *primary circuit* contains the source v_s and the *secondary circuit* contains the load V_L .

By using mesh analysis on each of the two loops, we can simplify the circuit to an equivalent circuit with a V_s in series with a reflected impedance Z_{in} :

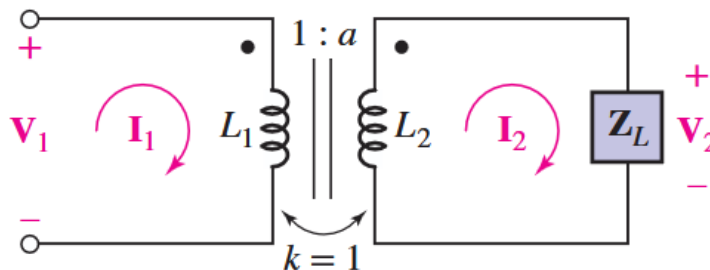
$$\frac{V_s}{I_1} = Z_{in} = Z_{11} - \frac{(j\omega)^2 M^2}{Z_{22}}$$

Where Z_{11} and Z_{22} are the self-impedances of the primary and secondary circuits, respectively.

$$Z_{11} = R_1 + j\omega L_1$$

$$Z_{22} = R_2 + j\omega L_2 + Z_L$$

- Ideal Transformer:



In an ideal transformer, the coupling coefficient $k = 1$ and the mutual inductance $M = \sqrt{L_1 L_2}$.

Turns Ratio:

$$a^2 = \frac{L_2}{L_1} = \frac{N_2^2}{N_1^2}$$

Using turns ratio, we can have the following relationships:

$$\frac{1}{a} = \frac{I_2}{I_1}$$

$$a = \frac{V_2}{V_1}$$

$$a^2 = \frac{\mathbf{Z_L}}{\mathbf{Z_{in}}}$$